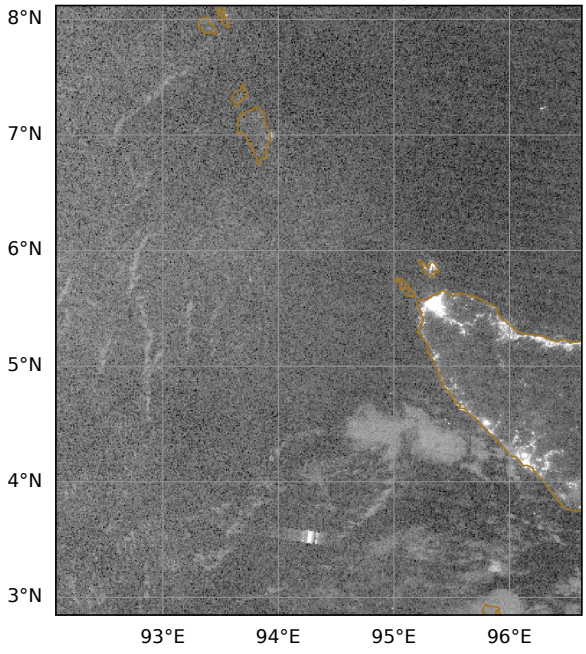
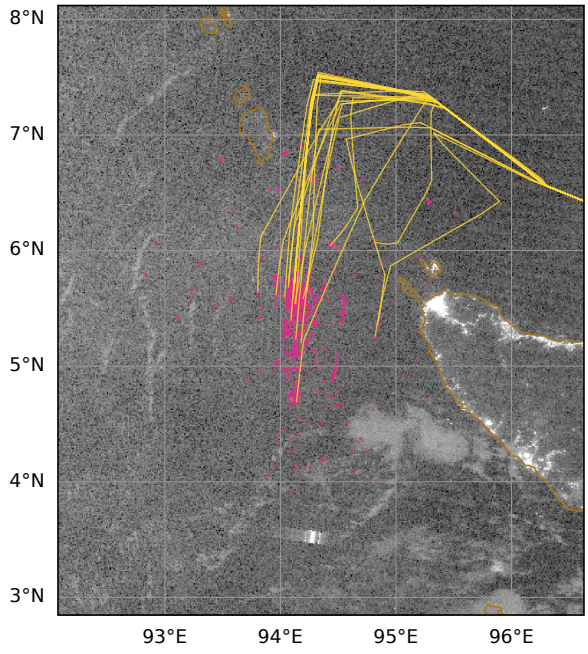
Suomi NPP VIIRS Day/Night Band, 18:55 UTC: moonless, so contrails are not assessable

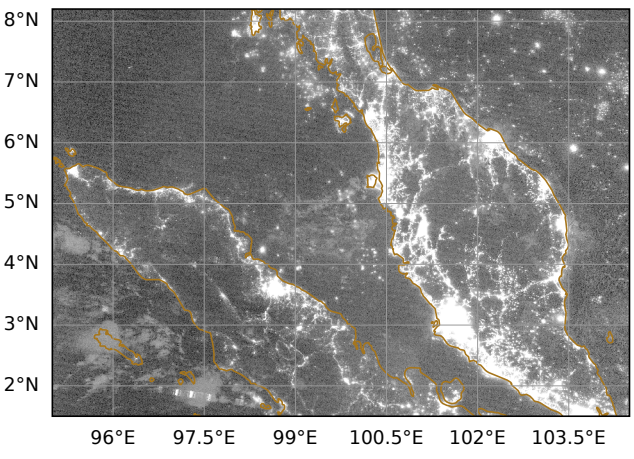
Likely aircraft area: image only



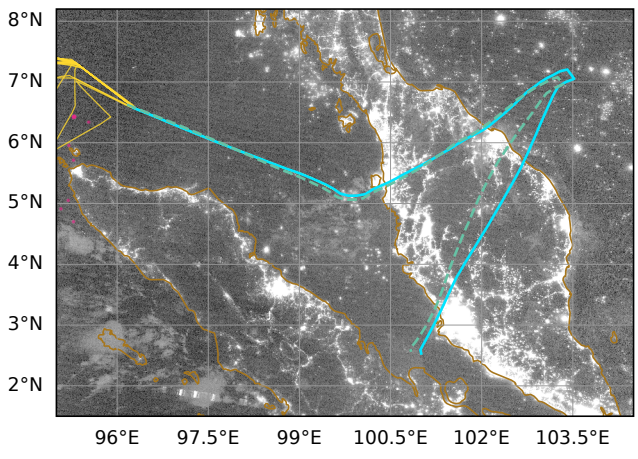
Likely aircraft area: with overlays

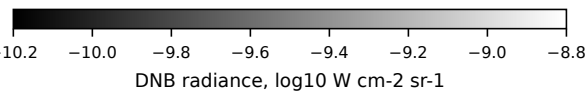


Radar-tracked flight area: image only



Radar-tracked flight area: with overlays





- likely aircraft position at the scan time (2,000 posterior samples)
- where a contrail up to 60 min old would lie (35 posterior routes, FL350 wind)
- where a contrail from the radar-tracked flight would lie, FL350 wind
- same, FL400 wind

Plain note: low-light picture from Suomi NPP at 18:55 UTC with no moonlight. Bright spots are towns and fishing boats. A contrail would need moonlight to show, so this picture cannot test for one.

Technical note: VIIRS Day/Night Band, Collection 2 (VNP02DNB.A2014066.1854.002, VNP03DNB.A2014066.1854.002; LAADS); pixel ~0.74 km across the swath; lunar zenith >= 109.4 deg everywhere in the scene (Moon below the horizon), Moon 42 % illuminated; Sun -75 deg at the aircraft. Scene lit only by airglow. Posterior: run J, BTO+BFO case (waypoints 0.9982, free 0.0013, reproduced-radar 0.0003, descent-climb 0.0002; 148,000 routes, linear interpolation between 600 s vertices); radar-tracked flight 16:42:27-18:22:12 from the Geoscience Australia digital military radar track. Positions use the granule's own scan time at each position and a parallax shift for FL350 (away from the sub-satellite point by 10.67 km x tan(view zenith)). Contrail loci use no contrail spreading; drift = ERA5 wind at FL350 (238.4 hPa) or FL400 (187.5 hPa) x contrail age..